Calculus II

Solution to Assignment 11

Part I. Regular Problems

1. $D_n = [0, n] \times [0, 1]$ is an increasing sequence of bounded rectangles whose union is D.

   (a) $\int_{D_n} \mathrm{e}^{-x}\, \mathrm{d}x\, \mathrm{d}y = \int_0^1 (-\mathrm{e}^{-n} + 1)\, \mathrm{d}y \to 1$ so $\int_D \mathrm{e}^{-x}\, \mathrm{d}x\, \mathrm{d}y = 1$.

   (b) $\int_{D_n} \mathrm{e}^{-x\sqrt{y}}\, \mathrm{d}x\, \mathrm{d}y = \int_0^1 \frac{-\mathrm{e}^{-n\sqrt{y}}+1}{\sqrt{y}}\, \mathrm{d}y$. Set $y = t^2$, then this integral is equal to

   $$\int_0^1 \left(-\mathrm{e}^{-nt} + 1\right) 2\, \mathrm{d}t = \frac{\mathrm{e}^{-n} - 1}{n} 2 + 2 \to 2$$

   $$\text{so } \int_D \mathrm{e}^{-x\sqrt{y}}\, \mathrm{d}x\, \mathrm{d}y = 2$$

   (c) In each sub-rectangle where $\frac{k-1}{n} \le y \le \frac{k}{n}$ you have $\mathrm{e}^{-xy} \ge \mathrm{e}^{-xk/n}$. Then

   $$\int_0^n \int_{\frac{k-1}{n}}^{\frac{k}{n}} \mathrm{e}^{-xy}\, \mathrm{d}y\, \mathrm{d}x \ge \int_0^n \frac{1}{n} \mathrm{e}^{-xk/n}\, \mathrm{d}x = \frac{1}{k}\left(1 - \mathrm{e}^{-k}\right) > \frac{1}{2k}$$

   Adding these, $\int_{D_n} \mathrm{e}^{-xy}\, \mathrm{d}A > \frac{1}{2}\left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}\right)$, which diverges. So $\int_D \mathrm{e}^{-xy}\, \mathrm{d}x\, \mathrm{d}y$ does not exist.

2. Let U be the open unit disk in $\mathbb{R}^2$ centered at the origin. $D_n = \{(x, y) \in \mathbb{R}^2 : 1 \le \sqrt{x^2 + y^2} \le n\}$ is an increasing sequence of bounded rectangles whose union is $\mathbb{R}^2 \setminus U$. Convert to polar coordinates: $r = \sqrt{x^2 + y^2}$, and change of variables gives us $dx\, dy = r\, dr\, d\theta$.

   (a) $\log r$:

   - The integral on D_n:

   $$\int_{r=1}^n \int_{\theta=0}^{2\pi} \log r \cdot r\, d\theta\, dr = 2\pi \int_1^n \log r \cdot r\, dr.$$

- Using integration by parts, the above integral diverges as $r \to \infty$. Hence, $\log r$ is **not integrable**.

(b) r^{-2}:

- The integral on D_n:

$$\int_{r=1}^{n} \int_{\theta=0}^{2\pi} r^{-2} \cdot r \, d\theta \, dr = 2\pi \int_{1}^{n} r^{-1} \, dr \to \infty$$

as $n \to \infty$, so r^{-2} is **not integrable**.

(c) $r^{-2.1}$:

- The integral on D_n:

$$\int_{r=1}^{n} \int_{\theta=0}^{2\pi} r^{-2.1} \cdot r \, d\theta \, dr = 2\pi \int_{1}^{n} r^{-1.1} \, dr.$$

- Since $-1.1 < -1$, the integral converges as $n \to \infty$. Therefore, $r^{-2.1}$ is **integrable**.

(d) r^{-3}:

- The integral on D_n:

$$\int_{r=1}^{n} \int_{\theta=0}^{2\pi} r^{-3} \cdot r \, d\theta \, dr = 2\pi \int_{1}^{n} r^{-2} \, dr.$$

- This integral converges as $n \to \infty$. Thus, r^{-3} is **integrable**.

3. (a)

$$\int_0^1 \int_1^2 \int_2^3 \cos[\pi(x+y+z)] \, dx \, dy \, dz$$

- First, compute the inner integral with respect to x:

$$\int_2^3 \cos[\pi(x+y+z)] \, dx = \frac{\sin[\pi(x+y+z)]}{\pi} \Bigg|_2^3.$$

- The sine terms can be simplified due to periodicity:

$$\sin[\pi(3+y+z)] - \sin[\pi(2+y+z)] = -2\sin[\pi(2+y+z)] = -2\sin[\pi(y+z)].$$

- Then we compute the inner integral with respect to y:

$$\int_1^2 \frac{-2}{\pi} \sin[\pi(y+z)] \, dy = \frac{2}{\pi^2} \cos[\pi(y+z)] \Bigg|_1^2$$

- The cosine terms can be simplified due to periodicity:

$$\cos[\pi(2+z)] - \cos[\pi(1+z)] = 2\cos[\pi z]$$

- We compute the integral with respect to z:

$$\int_0^1 \frac{4}{\pi^2}\cos[\pi z]\,dz = \frac{4}{\pi^3}\sin[\pi z]\Big|_0^1 = \frac{4}{\pi^3}(\sin[\pi] - \sin[0]) = 0$$

- Therefore, the integral evaluates to:

$$\boxed{0}.$$

(b)

$$\int_0^1 \int_0^x \int_0^y (y + xz)\,dz\,dy\,dx$$

- Start with the inner integral with respect to z:

$$\int_0^y (y + xz)\,dz = y \cdot z + \frac{xz^2}{2}\Big|_0^y = y^2 + \frac{xy^2}{2}.$$

- Now integrate with respect to y:

$$\int_0^x \left(y^2 + \frac{xy^2}{2}\right) dy = \frac{y^3}{3} + \frac{xy^3}{6}\Big|_0^x = \frac{x^3}{3} + \frac{x^4}{6}.$$

- Finally, integrate with respect to x:

$$\int_0^1 \left(\frac{x^3}{3} + \frac{x^4}{6}\right) dx = \frac{x^4}{12} + \frac{x^5}{30}\Big|_0^1 = \frac{1}{12} + \frac{1}{30} = \frac{7}{60}.$$

- Therefore, the integral evaluates to:

$$\boxed{\frac{7}{60}}.$$

(c)

$$\int_0^2 \int_0^x \int_0^{x+y} dz\,dy\,dx$$

- Start with the inner integral with respect to z:

$$\int_0^{x+y} dz = x + y.$$

- Now integrate with respect to y:

$$\int_0^x (x+y)\,dy = x\cdot y + \frac{y^2}{2}\bigg|_0^x = x^2 + \frac{x^2}{2} = \frac{3x^2}{2}.$$

- Finally, integrate with respect to x:

$$\int_0^2 \frac{3x^2}{2}\,dx = \frac{3}{2}\cdot\frac{x^3}{3}\bigg|_0^2 = \frac{3}{2}\cdot\frac{8}{3} = 4.$$

- Therefore, the integral evaluates to:

$$\boxed{4}.$$

4. Here $E = \{(x,y,z) \mid 0 \le x \le 1, 0 \le y \le \sqrt{x}, 0 \le z \le 1+x+y\}$, so

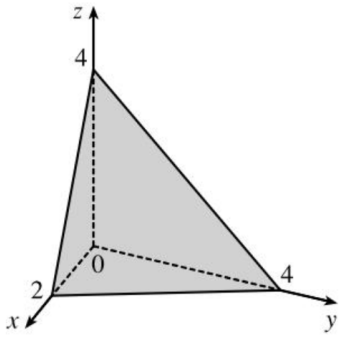
$$\begin{aligned}\iiint_E 6xy dV &= \int_0^1\int_0^{\sqrt{x}}\int_0^{1+x+y} 6xy\,dz\,dy\,dx = \int_0^1\int_0^{\sqrt{x}} [6xyz]_{z=0}^{z=1+x+y}\,dy\,dx \\ &= \int_0^1\int_0^{\sqrt{x}} 6xy(1+x+y)\,dy\,dx = \int_0^1 \left[3xy^2 + 3x^2y^2 + 2xy^3\right]_{y=0}^{y=\sqrt{x}}\,dx \\ &= \int_0^1 \left(3x^2 + 3x^3 + 2x^{5/2}\right)dx = \left[x^3 + \frac{3}{4}x^4 + \frac{4}{7}x^{7/2}\right]_0^1 = \frac{65}{28}\end{aligned}$$

5. The plane $2x + y + z = 4$ intersects the xy-plane when

$$2x + y + 0 = 4 \Rightarrow y = 4 - 2x,$$

so

$$E = \{(x,y,z) \mid 0 \le x \le 2, 0 \le y \le 4-2x, 0 \le z \le 4-2x-y\}$$



And

$$\begin{aligned} V &= \int_0^2 \int_0^{4-2x} \int_0^{4-2x-y} dzdydx = \int_0^2 \int_0^{4-2x} (4-2x-y)dydx \\ &= \int_0^2 \left[4y - 2xy - \frac{1}{2}y^2\right]_{y=0}^{y=4-2x} dx \\ &= \int_0^2 \left[4(4-2x) - 2x(4-2x) - \frac{1}{2}(4-2x)^2\right] dx \\ &= \int_0^2 \left(2x^2 - 8x + 8\right) dx = \left[\frac{2}{3}x^3 - 4x^2 + 8x\right]_0^2 = \frac{16}{3} \end{aligned}$$

6.
$$\begin{aligned} V(E) = L^3 \Rightarrow f_{\text{ave}} &= \frac{1}{L^3}\int_0^L \int_0^L \int_0^L xyzdxdydz = \frac{1}{L^3}\int_0^L xdx \int_0^L ydy \int_0^L zdz \\ &= \frac{1}{L^3}\left[\frac{x^2}{2}\right]_0^L \left[\frac{y^2}{2}\right]_0^L \left[\frac{z^2}{2}\right]_0^L = \frac{1}{L^3}\frac{L^2}{2}\frac{L^2}{2}\frac{L^2}{2} = \frac{L^3}{8} \end{aligned}$$

7. (a) In cylindrical coordinates, E is given by $\{(r,\theta,z) \mid 0 \le \theta \le 2\pi, 0 \le r \le 4, -5 \le z \le 4\}$. So

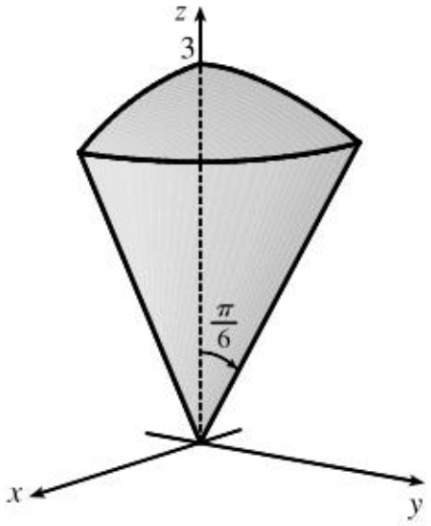
$$\begin{aligned} \iiint_E \sqrt{x^2+y^2}dV &= \int_0^{2\pi}\int_0^4 \int_{-5}^4 \sqrt{r^2} r dz dr d\theta = \int_0^{2\pi} d\theta \int_0^4 r^2 dr \int_{-5}^4 dz \\ &= [\theta]_0^{2\pi}\left[\frac{1}{3}r^3\right]_0^4 [z]_{-5}^4 = (2\pi)\left(\frac{64}{3}\right)(9) = 384\pi \end{aligned}$$

(b) In cylindrical coordinates, E is bounded by the cylinder $r = 1$, the plane $z = 0$, and the cone $z = 2r$. So

$$E = \{(r,\theta,z) \mid 0 \le \theta \le 2\pi, 0 \le r \le 1, 0 \le z \le 2r\}.$$

$$\begin{aligned} \iiint_E x^2 dV &= \int_0^{2\pi}\int_0^1 \int_0^{2r} r^2\cos^2\theta r dz dr d\theta = \int_0^{2\pi}\int_0^1 \left[r^3\cos^2\theta z\right]_{z=0}^{z=2r} dr d\theta \\ &= \int_0^{2\pi}\int_0^1 2r^4\cos^2\theta dr d\theta \\ &= \int_0^{2\pi}\left[\frac{2}{5}r^5\cos^2\theta\right]_{r=0}^{r=1} d\theta = \frac{2}{5}\int_0^{2\pi}\cos^2\theta d\theta = \frac{2}{5}\int_0^{2\pi}\frac{1}{2}(1+\cos 2\theta)d\theta \\ &= \frac{1}{5}\left[\theta + \frac{1}{2}\sin 2\theta\right]_0^{2\pi} = \frac{2\pi}{5} \end{aligned}$$

8. The region of integration is given in spherical coordinates by $E = \{(\rho,\theta,\phi) \mid 0 \le \rho \le 3, 0 \le \theta \le \pi/2, 0 \le \phi \le \pi/6\}$.



This represents the solid region in the first octant bounded above by the sphere $\rho = 3$ and below by the cone $\phi = \pi/6$.

$$\begin{aligned}\int_0^{\pi/6}\int_0^{\pi/2}\int_0^3 \rho^2 \sin\phi \, d\rho d\theta d\phi &= \int_0^{\pi/6} \sin\phi d\phi \int_0^{\pi/2} d\theta \int_0^3 \rho^2 d\rho \\ &= [-\cos\phi]_0^{\pi/6}[\theta]_0^{\pi/2}\left[\frac{1}{3}\rho^3\right]_0^3 \\ &= \left(1 - \frac{\sqrt{3}}{2}\right)\left(\frac{\pi}{2}\right)(9) = \frac{9\pi}{4}(2-\sqrt{3})\end{aligned}$$

9. In spherical coordinates, the sphere $x^2+y^2+z^2 = 4$ is equivalent to $\rho = 2$ and the cone $z = \sqrt{x^2+y^2}$ is represented by $\phi = \frac{\pi}{4}$ (as in Example 4). Thus, the solid is given by $\left\{(\rho,\theta,\phi) \mid 0 \le \rho \le 2, 0 \le \theta \le 2\pi, \frac{\pi}{4} \le \phi \le \frac{\pi}{2}\right\}$ and

$$\begin{aligned}V &= \int_{\pi/4}^{\pi/2}\int_0^{2\pi}\int_0^2 \rho^2 \sin\phi \, d\rho d\theta d\phi = \int_{\pi/4}^{\pi/2} \sin\phi d\phi \int_0^{2\pi} d\theta \int_0^2 \rho^2 d\rho \\ &= [-\cos\phi]_{\pi/4}^{\pi/2}[\theta]_0^{2\pi}\left[\frac{1}{3}\rho^3\right]_0^2 = \left(\frac{\sqrt{2}}{2}\right)(2\pi)\left(\frac{8}{3}\right) = \frac{8\sqrt{2}\pi}{3}\end{aligned}$$

10. In spherical coordinates $z = \sqrt{x^2+y^2}$ becomes $\phi = \frac{\pi}{4}$. Then

$$V = \int_0^{2\pi} \int_0^{\pi/4} \int_0^1 \rho^2 \sin\phi d\rho d\phi d\theta = \int_0^{2\pi} d\theta \int_0^{\pi/4} \sin\phi d\phi \int_0^1 \rho^2 d\rho$$
$$= 2\pi \left(-\frac{\sqrt{2}}{2} + 1 \right) \left(\frac{1}{3} \right) = \frac{1}{3}\pi(2 - \sqrt{2})$$

Part II. Group Problems

1. (a) $\iiint_B x^2 \, dx \, dy \, dz$, where $B = [0,1] \times [0,1] \times [0,1]$
We need to evaluate the integral:

$$\iiint_B x^2 \, dx \, dy \, dz$$

By Fubini's theorem, the integral can be calculated as follows:

$$\int_0^1 \int_0^1 \int_0^1 x^2 \, dx \, dy \, dz$$

Start by integrating with respect to x:

$$\int_0^1 x^2 \, dx = \frac{x^3}{3}\Big|_0^1 = \frac{1}{3}$$

Since the integrand no longer depends on y and z, we can now integrate with respect to y and z:

$$\int_0^1 \int_0^1 1 \, dy \, dz = 1$$

Thus, the result for part (a) is:

$$\boxed{\frac{1}{3}}$$

(b) $\iiint_B e^{-xy} y \, dx \, dy \, dz$, where $B = [0,1] \times [0,1] \times [0,1]$
We now evaluate the integral:

$$\iiint_B e^{-xy} y \, dx \, dy \, dz$$

By Fubini's theorem, the integral can be calculated as follows:

$$\int_0^1 \int_0^1 \int_0^1 e^{-xy} y \, dx \, dy \, dz$$

We start by integrating with respect to x. Note that y is treated as a constant in this integration:

$$\int_0^1 y e^{-xy} \, dx = y \left[\frac{-e^{-xy}}{y} \right]_0^1 = 1 - e^{-y}$$

Now, integrate with respect to y:

$$\int_0^1 (1 - e^{-y}) \, dy$$

This integral splits into two simpler integrals:

$$\int_0^1 1 \, dy - \int_0^1 e^{-y} \, dy$$

The first integral is straightforward:

$$\int_0^1 1 \, dy = 1$$

For the second integral:

$$\int_0^1 e^{-y} \, dy = \left[-e^{-y} \right]_0^1 = 1 - \frac{1}{e}$$

Thus, the result of the y-integral is:

$$1 - \left(1 - \frac{1}{e} \right) = \frac{1}{e}$$

Now, integrate with respect to z, which gives:

$$\int_0^1 \frac{1}{e} \, dz = \frac{1}{e}$$

Thus, the result for part (b) is:

$$\boxed{\frac{1}{e}}$$

(c) $\iiint_B (2x + 3y + z)\,dx\,dy\,dz$, where $B = [0, 2] \times [-1, 1] \times [0, 1]$
We need to evaluate the integral:

$$\iiint_B (2x + 3y + z)\,dx\,dy\,dz$$

By Fubini's theorem, the integral can be calculated as follows:

$$\int_0^1 \int_{-1}^1 \int_0^2 (2x + 3y + z)\,dx\,dy\,dz$$

We break the integral into three parts:

$$\int_0^1 \int_{-1}^1 \int_0^2 2x\,dx\,dy\,dz + \int_0^1 \int_{-1}^1 \int_0^2 3y\,dx\,dy\,dz + \int_0^1 \int_{-1}^1 \int_0^2 z\,dx\,dy\,dz$$

First part: $\int_0^2 2x\,dx$

$$\int_0^2 2x\,dx = \left[x^2\right]_0^2 = 4$$

Thus, the first part of the integral becomes:

$$\int_0^1 \int_{-1}^1 4\,dy\,dz = 4 \cdot 2 \cdot 1 = 8$$

Second part: $\int_0^2 3y\,dx$
Since y is treated as a constant, we integrate:

$$\int_0^2 3y\,dx = 3y \cdot 2 = 6y$$

Now, integrate with respect to y and z:

$$\int_0^1 \left(\int_{-1}^1 6y\,dy \right) dz = \int_0^1 0\,dz = 0$$

Third part: $\int_0^2 z\,dx$

$$\int_0^2 z\,dx = 2z$$

Now, integrate with respect to z:

$$\int_0^1 \left(\int_{-1}^1 2z\,dy \right) dz = \int_0^1 4z\,dz = \left[2z^2\right]_0^1 = 2$$

Final Result:
Combining the results of the three parts, we get:

$$8 + 0 + 2 = 10$$

Thus, the result for part (c) is:

$$\boxed{10}$$